

DULE ABERA BULTI

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Professional Summary

AI Engineer and Data Scientist specializing in computer vision and crisis intelligence systems, with proven experience building real-time AI solutions achieving up to 94.7% mAP. Experienced in deploying YOLO-based models, multimodal analytics, and interpretable ML systems for agriculture, surveillance, and humanitarian applications. Published researcher (IEEE AGERS 2025, ICCED 2025) with a strong focus on translating AI research into scalable, real-world impact.

Education

Master's Degree, Data Science - Universitas Nusa Putra, Indonesia December 2024 – Present

- **Thesis:** Real-Time Crisis Recognition and Response System for Protest Violence Detection
- **Focus:** Computer Vision, Human Activity Recognition, Crisis Intelligence Systems, and Multimodal Learning.

Bachelor of Electrical and Computer Engineering (Computer Stream), Haramaya University - Ethiopia (2018 - 2023)

- Focused on Computer Stream within Electrical and Computer Engineering
- Developed solid foundation in computer science principles and electrical engineering concepts

Work Experience

Graduate Trainee - Cooperative Bank of Oromia, Addis Ababa, Ethiopia March 2024 – December 2024

- Assisted and developed in deploying an ML project used by internal analytics teams
- Contributed to the deployment of a machine learning model used by internal analytics teams, improving data-driven decision-making efficiency.
- Collaborated with cross-functional teams to integrate AI solutions into existing banking infrastructure

AI/ML Research Intern - Ethiopian Artificial Intelligence (AI) Company - Addis Ababa, Ethiopia October 2022 - January 2023

- Collaborated with AI development teams to enhance AI model performance
- Conducted systematic hyperparameter optimization and architecture refinement experiments
- Contributed to research initiatives focused on improving model efficiency and interpretability

Projects

Oil Palm Tree Health Detection System | YOLOv8, YOLOv12, Streamlit, UAV Imaging | Published (IEEE AGERS 2025)

- Developed automated detection system for oil palm tree health monitoring with **94.7% mAP**
- Performed tree detection, counting, and multiclass health classification (Healthy, Yellowing, Small, Dead)
- Deployed web application using Streamlit for real-time field monitoring

Real-Time Crisis Recognition and Response System for Protest Violence Detection | YOLO, OpenCV | Published (ICCED 2025)

- Architected multimodal AI framework combining object detection, 17-keypoint pose estimation, and behavioral scoring
- Analyzed joint angles motion and hand-object proximity for violence classification (Throwing, Fighting, Pushing)
- Generated timestamped crisis alerts with annotated keyframes for law enforcement situational awareness

Conflict-Induced Food Crisis Prediction

- Developed a machine learning system to forecast food crises using conflict, climate, and historical food security data.
- Applied data fusion, feature engineering, and temporal modeling to capture crisis patterns and trends.
- Implemented a tuned XGBoost model with explainability (SHAP) for accurate and interpretable predictions.

Fake Image Detection

- Built a deep learning pipeline to detect AI-generated images using pixel-level forensic features.
- Utilized CNNs and transfer learning (EfficientNet) to improve classification performance.
- Incorporated explainability techniques (Grad-CAM) to validate model focus on meaningful visual artifacts.

Skills

- **Strong work ethic, Teamwork, Problem-solving, Critical thinking, Communication,**
- **ML/AI Frameworks:** TensorFlow, PyTorch, OpenCV, YOLO, Transfer Learning, Model Optimization& Deployment
- **Data Analytics:** Pandas, NumPy, Power BI, Seaborn, Matplotlib
- **Other:** Streamlit, Git, Docker

Certifications

- **Building AI Projects: Master Machine Learning & Deep Learning**
- **The Complete MySQL Bootcamp: Go from Beginner to Expert ,**
- **Artificial Intelligence Foundations: Machine Learning,**
- **Basics of Cloud Computing: Fundamentals of Cloud Computing**
- **Power BI Mastery: Zero to Hero Data Skills**
- **KAIM 10 Academy**

Publications

- **Oil Palm Tree Health Detection System Using YOLO-based Deep Learning Models and UAV Imagery.**
IEEE AGERS 2025. (First Author) DOI: [10.1109/AGERS67633.2025.11446373](https://doi.org/10.1109/AGERS67633.2025.11446373)
- **Real-Time Crisis Recognition and Response System for Protest Violence Detection.**
ICCED 2025. (First Author) DOI: [10.1109/ICCED68324.2025.11324702](https://doi.org/10.1109/ICCED68324.2025.11324702)
- **Evaluation of YOLOv11 and RF-DETR Architectures for Detection of Healthy Oil Palm Trees.**
IC2IE 2025. (Co-Author) DOI: [10.1109/IC2IE67206.2025.11283417](https://doi.org/10.1109/IC2IE67206.2025.11283417)
- **The Impact of Online Learning on Student Performance Using a Causal Inference Approach.**
ICISIT 2026. (Co-Author) DOI: [10.1007/978-3-032-14750-9_14](https://doi.org/10.1007/978-3-032-14750-9_14)
- **Analysis of Web User Behavior for Bandung City."**
ICISIT 2026. (Co-Author) DOI: [10.1007/978-3-032-14750-9_13](https://doi.org/10.1007/978-3-032-14750-9_13)

Research interests

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| • Computer Vision | • Precision Agriculture |
| • Deep Learning | • Remote Sensing and UAV-based Analytics |
| • Artificial Intelligence | • Crisis Intelligence and Humanitarian AI |
| • Machine Learning | • Food Security Prediction |
| • Intelligent Surveillance Systems | • Multimodal Learning Systems |
| • Human Activity Recognition | • Data Science and Predictive Analytics |